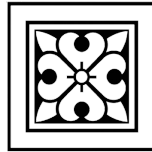


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BIOGRAPHY



Born on March 21, 1768, in Auxerre, France. Joseph Fourier was the son of a tailor, but he would rise to prominence as one of the most influential mathematicians and physicists of his time. Fourier lost his parents at an early age and was subsequently raised by a Benedictine abbey, where he received his early education. Showing exceptional promise in mathematics, he pursued his studies at the École Royale Militaire of Auxerre.

In 1787, Fourier began teaching mathematics at the Benedictine college in his hometown. His academic career was briefly interrupted by the French Revolution, during which he became involved in political activities. He was appointed as a commissioner for the National Convention, a role in which he narrowly escaped execution during the Reign of Terror.

After the Revolution, Fourier returned to academia and continued his work in mathematics. He joined the newly established École Normale Supérieure in Paris in 1795, where he studied under renowned mathematicians such as Joseph-Louis Lagrange and Pierre-Simon Laplace.

Fourier's most significant contribution to science came during his time in Egypt with Napoleon Bonaparte's expedition. Appointed as the secretary of the Institut d'Égypte, he conducted important research in heat transfer. This work culminated in his seminal publication, *Théorie analytique de la chaleur* (The Analytical Theory of Heat) in 1822. In this treatise, Fourier introduced the concept of Fourier series, a mathematical tool that allows the decomposition of functions into sine and cosine components. word ■ word